CODING FOR DATA SCIENCE AND ANALYTICS

Exercise Session 5: Lists, Tuples, and Dictionaries

Exercise 1 — Finding All Indices in a List

Python's `.index(target)` method only returns the index of the **first** occurrence of an element.

```
numbers = [4, 3, 5, 6, 4, 1, 4, 9]
```

Goal

Write a function that returns a new list containing **all** the indices where `target` appears (e.g., `[0, 4, 6]` for `target = 4`).

Two Approaches

1. **Version 1 (Using `.index()`):** Use a `for` loop and the `lst.index(target, start)` method.
2. **Version 2 (Using `enumerate()`):** Rewrite the function using `enumerate(lst)` without calling `.index()`.

Exercise 2 — Simulating a Stack: Text Editor "Undo"

Using `.append()` and `.pop()`, lists can simulate a **Stack** data structure (Last-In, First-Out). Let's build the "Undo" feature of a text editor!

Goal

Start with an empty list `document = []`. Write a `while` loop (controlled by a sentinel variable) that continuously asks the user for a word or command:

- **Any normal word:** add it to the end of `document`.
- **`"undo"`:** remove the last inserted word.
- **`"print"`:** display the current words in `document`.
- **`"exit"`:** terminate the loop and print the final document.

Exercise 3 — Dictionaries: Corporate Acronym Expander

Let's build a text expander using a dictionary that maps business acronyms to their full meaning:

```
acronyms = {
    "asap": "as soon as possible",
    "roi": "return on investment",
    "kpi": "key performance indicator",
    "b2b": "business to business",
    "ceo": "chief executive officer"
}
```

Tasks

1. Add a new acronym `"ai":"artificial intelligence"` and remove `"b2b"`.
2. Rebuild the sentence "The ceo wants the kpi and roi asap" by replacing any known acronym with its full text.

Exercise 4 — Credit Card Statement

You have a credit card statement stored as a **list of tuples**. Each tuple represents a transaction `(category, amount)`:

```
transactions = [
    ("Food", 45), ("Transport", 20), ("Food", 30),
    ("Entertainment", 60), ("Transport", 15),
    ("Food", 25), ("Entertainment", 40), ("Shopping", 80)
]
```

Tasks (Mixed Data Structures)

1. Loop through the list of tuples using **sequence unpacking**.
2. Accumulate the total amount spent per category inside a **dictionary**.
3. Print a summary table of the expenses per category alongside an ASCII bar chart made of asterisks $*$ (**1 asterisk for every 10€ spent**).

Exercise 5 — 2D Lists: Cinema Seating Map

A cinema hall is represented as a 2D list (4 rows × 5 seats), where `'O'` means **Available** and `'X'` means **Occupied**:

```
cinema = [
    ['X', 'O', 'O', 'O', 'X'],
    ['X', 'X', 'O', 'X', 'X'],
    ['O', 'O', 'O', 'O', 'X'],
    ['X', 'X', 'X', 'O', 'O']
]
```

Tasks

1. Use nested `enumerate()` loops to print the seating map nicely with Row and Seat numbers, and count the total number of available seats (`'O'`).
2. Find and print which row has the **highest number** of available seats.
3. Book Row 0, Seat 2 by changing its value and verify that the map updated.